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# Enabling Effective Metamorphic-Relation Generation by Novice Testers: A Pilot Study

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***Abstract*—This paper presents a pilot study that examines the capacity of novice testers to generate Metamorphic Relations (MRs) for autonomous driving systems (ADSs), specifically focusing on parking functions. By comparing MRs generated by human participants with those generated by artificial intelligence (AI), we seek to understand the variances in quality, particularly in terms of correctness, applicability, novelty, and utility. Our findings indicate that despite receiving only minimal training, human participants were capable of producing MRs with a wide range of effectiveness. Notably, humans exhibited a potential for creative thinking, contrasting with AI's ability to generate MRs that adhere closely to technical and applicability standards. The study underscores the need for improved educational strategies aimed at enhancing the quality and confidence of MRs produced by humans. Future research directions will explore the optimization of training approaches, particularly within a constrained timeframe to create a positive learning experience and maintain participant engagement, to fully harness the creative capabilities of human learners in the context of ADS testing.**

***Index Terms*—Metamorphic testing, autonomous driving system, metamorphic relation, driving scenarios, large language models, artificial intelligence**

## I. Introduction

The development and validation of autonomous driving systems (ADSs) requires innovative testing approaches due to their complex and dynamic nature [1]. Among these, Metamorphic Testing (MT) has become popular for its effectiveness in addressing the challenge of test oracle uncertainty in ADS [2]. MT leverages Metamorphic Relations (MRs), which are relations that should hold among the inputs and related outputs of multiple executions of the program under test [3]. MRs serve as the cornerstone of MT by enabling testers to alleviate the oracle problem—the difficulty of knowing the correct outcome within a reasonable amount of time—which is particularly obvious in ADS due to the infinite range of possible driving situations [2].

As MT continues to gain popularity, finding effective methods to educate students, professional software engineers, testers, and end-users about it has emerged as a critical issue [3]. One of the challenges in educating testers to generate MRs, especially for complex systems such as ADSs, arises from the constraints of limited time and resources [4]. This difficulty is particularly acute in scenarios such as guest lectures or workshops where participants, often novice testers, are introduced to the complexities of MR generation in a condensed timeframe [5]. For example, in a university setting, a guest introduction lecture might offer only a single session to cover the principles of MRs, leaving little room for in-depth exploration or hands-on practice. Similarly, project teams facing urgent deadlines may need to quickly upskill members in MR generation to meet testing requirements, operating under the dual pressures of time scarcity and resource limitations [6]. These situations highlight the need for efficient and effective training methods that can rapidly equip participants with the necessary skills to generate meaningful MRs, despite the challenge of navigating these constraints.

In this paper, we present a pilot study that focuses on the potential of novice testers, with little to no prior experience in software testing or ADSs, to generate MRs for a critical ADS function: automated parking [7]. The initiative is based on the hypothesis that even individuals new to the field can contribute valuable insights, provided they are given appropriate guidance [4]. Our investigation centers on assessing the MRs with an existing evaluation framework from an earlier study [8], which ranges across multiple dimensions: correctness, applicability, novelty, clarity, relevance to safety, and overall usefulness. These criteria are critical in determining the quality of MRs for ADSs. Given the limited exposure of participants to MR generation and ADS functionalities, this study also evaluates the educational strategies employed, focusing on their ability to teach essential skills and foster confidence among novice testers. The findings reveal that correctness proved challenging, with issues in clearly depicting the actions and events of traffic participants in the scenario, as well as weak connections between the source and follow-up test scenarios within the same MR. Conversely, applicability showed promise, indicating some MRs could be useful in practical testing. However, clarity, safety relevance, and computational feasibility were areas of weakness, with many MRs failing to provide clear, safety-focused, and realistically testable scenarios. Despite these challenges, certain innovative MRs demonstrated potential, indicating promising performance in revealing bugs.

To further explore the effectiveness of the MRs, we compared them against others generated by ChatGPT in our

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earlier study [8], which also explored generating MRs for the automated parking function, evaluating them under the same criteria. The findings reveal that while AI-generated MRs excel in accuracy and clarity, human novice testers demonstrate the strength for innovative thought. Future research directions will explore the optimization of training approaches, particularly within a constrained timeframe to create a positive learning experience and maintain participant engagement, to fully harness the creative capabilities of human learners in the context of ADS testing.

The rest of the paper is organized as follows: Section II introduces the background of the study. Section III describes the design of the study. Section IV presents the evaluation results of the MRs, followed by Section V that compares the quality of human-generated MRs with AI-generated ones, as well as the limitations and potential future work of this study. Section VI concludes the paper.

## II. Background

### A. Autonomous Driving Systems

Autonomous driving systems (ADSs) are systems designed to enable vehicles to navigate and operate without human intervention, using a complex array of sensors, cameras, artificial intelligence (AI), and machine learning algorithms [2]. ADSs can perceive their environment, make decisions, and execute maneuvers to reach a destination safely [1]. Typical hardware used by ADSs include LIDAR and radar for object detection, cameras for object recognition, and algorithms for decision-making [1]. Driving scenarios in the context of self-driving are critical for both the development and evaluation of these systems [9]. They encompass a wide range of situations that a vehicle might encounter on the road, including but not limited to navigating urban and rural environments, interacting with pedestrians and other vehicles, handling adverse weather conditions, and responding to unexpected obstacles or emergencies [9]. This includes a variety of weather conditions, the complexity of urban landscapes, and diverse traffic scenarios [10]. These scenarios are significant in training and testing the ADS algorithms.

### B. Metamorphic Testing

Metamorphic Testing (MT) has emerged as a powerful technique to alleviate the oracle problems where traditional testing methods may fall short [3]. As an end-user-friendly technique [11][12], MT makes use of Metamorphic Relations (MRs) to generate tests, where MRs are relations expected to hold among multiple inputs and the related outputs [13]. By identifying and exploiting these relations, MT can uncover defects that might not be revealed by conventional testing methods, making it especially useful in fields like machine learning, numerical computations, and software dealing with complex algorithms or systems where the exact output is inherently unpredictable [14].

Teaching how to identify/generate MRs is a critical step in leveraging the full potential of MT [14]. It involves a deep understanding of the SUT, including its expected behavior under various conditions and how different input parameters might influence the outcome. The process starts with identifying key properties of the software that remain invariant or that change predictably under certain transformations of the input [4]. Through hands-on exercises, case studies, and explorations of real-world examples, learners can develop the skills necessary to identify and apply MRs effectively [3].

## III. Methodology

### A. Recruitment of Participants

This pilot study recruited four male and two female students via email invitations. They were all novices in software testing, MT, and ADSs. All participants are currently undergraduate students.

### B. Study Design

The study was structured to introduce students to the concept and application of MRs within the context of ADSs. The design consisted of a one-hour session divided into two parts: a forty-minute lecture and a twenty-minute hands-on workshop.

The initial lecture aimed to equip students with the necessary knowledge to understand and generate MRs and scenarios related to ADS functions. It started with a broad overview of ADSs using real-life examples, then delved into the challenges of testing these intricate systems, introducing the concept of MT. Following a brief explanation of essential concepts, students were instructed on creating and generating driving scenarios and MRs. The lecture detailed the fundamental elements of driving scenarios and their utilization in the generation of MRs, supported by various examples [2][14][15]. Since the lecture only took forty minutes, only the necessary knowledge required to generate MRs for the workshop task was taught, aiming to maximise efficiency.

Following this, the workshop provided a hands-on experience where students were challenged to generate two MRs specifically for an ADS parking function, in which they were provided with a paragraph describing its typical functionality and usage to ensure their understanding of the SUT. Throughout the session, students were encouraged to ask questions related to the lecture content to deepen their understanding. However, to ensure the originality and unbiased nature of the MRs generated, they were restricted from requesting hands-on instructions on tasks or example answers in the workshop. This approach aimed to prevent instructor bias [16] from impacting the quality of the MRs produced.

The choice of the parking function as the focus for MR generation was strategic. The parking function, being a task closely related to the everyday driving experience, was considered to be more accessible and understandable for novice testers compared to more complex ADS functions such as



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TABLE I: Marking table for evaluating the quality of MRs [8]

| Criteria | Description | Score |
|---|---|---|
| Correctness | Does the MR accurately capture the intended behavior? | 0-5 |
| Applicability | Is the MR applicable to a wide range of inputs? | 0-5 |
| Novelty | Is the MR a new and original idea? | 0-5 |
| Clarity | Is the MR easy to understand and unambiguous? | 0-5 |
| Relevance to safety | Does the MR address a safety-critical aspect of the module? | 0-5 |
| Overall usefulness | How useful is the MR for testing the module? | 0-5 |
| Computational feasibility | Is it computationally feasible to apply the MR? | 0-5 |

Adaptive Cruise Control[1] (ACC) and Lane Keeping Assist[2] (LKA). Additionally, the relatively closed environment of a parking lot offered a more manageable setting for imagining and designing scenarios.

### C. Data Collection

Data collection involved gathering students' generated MRs and their responses on provided answer sheets. These sheets were structured to facilitate organized thought and clarity in the generation of MRs. Each sheet was divided into sections corresponding to the components of an MR, including the source scenario, input relation, follow-up scenario, output relation, and their combinations [13]. The purpose of structuring the answer sheets in this way was to guide the participant's thought process, taking them from comprehending the ADS parking function in a practical setting to conceiving relationships among various scenarios. By dividing the MR generation process into different parts, this structure attempted to reduce the difficulty of MR generation while taking into account the students' diverse academic backgrounds and degrees of skill.

After gathering the MRs, the students were encouraged to take a short self-evaluation survey to report their experience in learning and generating MRs. The self-evaluation surveys assessed students' ease with which they generated them and their confidence in their own MRs. The ease of MR generation ranged from 'Very Difficult' to 'Very Easy', indicating the students' perceived difficulty of the task and their need for assistance, as shown in Table II. Confidence levels ranged from 'Very Low' to 'Very High', reflecting students' trust in the quality and applicability of their MRs, as shown in Table III. These surveys aimed to evaluate the effectiveness of the instructional methods and the student's grasp of MR generation. The results of the survey are provided in Section V-B.

### D. Data Analysis

The analysis of the MRs and scenarios generated by the students employed the evaluation criteria derived from our previous study [8]. The evaluation was performed by the same group of evaluators, who are experts in the area. This evaluation framework, as shown in Table I, included several criteria: correctness, applicability, novelty, clarity, relevance to safety, overall usefulness, and computational feasibility. Correctness assessed the accuracy of the MRs, ensuring they accurately represented the intended ADS behavior. Applicability examined the MRs' adaptability to various inputs and situations. Novelty focused on the originality of the proposed MRs. Clarity evaluated the ease of understanding the MRs. The relevance to safety criterion looked at how the MRs addressed safety-critical aspects of the ADS. Overall usefulness gauged the MRs' practical value in testing and verification processes, and computational feasibility measured the ease with which the MRs could be implemented in practical scenarios. The analysis helps study the diversity and richness of the MRs generated by students. The results are presented in Section IV, and we also provide a comparison to the qualities of MRs generated by ChatGPT from our earlier study [8] in Section V-A, as both studies used the same set of evaluation criteria and the same target system to generate MRs (i.e., an ADS parking function).

## IV. Findings

The experiment aimed to assess the capability of novice testers to identify/generate MRs for ADS parking functions. Each of the six participating students was tasked with generating two MRs, resulting in a total of 12 MRs under evaluation. The MRs were assessed following the criteria in Table I, with each criterion rated on a scale of 1 to 5.

### A. Correctness

The average score for the correctness criterion was **2.9**, indicating that approximately half of the MRs encountered issues with correctness. The main issues revolve around the weak or absent connections between the source and follow-up scenarios, and unclear descriptions of certain elements of driving scenarios, like the precise actions of the self-driving car (the EGO vehicle) during events. For instance, the following is an example of the human-generated MR which has some problems:

> *"In the source scenario, at a crowded mall parking lot with only two available parking spots (one big and one small). The EGO car is small and can fit in both spots. Soon another big car (B) comes in, which can only fit in the big parking spot. And the EGO vehicle has not driven into the parking space. In the follow-up scenario, the EGO vehicle had better drive in the smaller spot so another car (B) can fit into the bigger one. The SUT can utilize all the space fully."*

[1]Adaptive Cruise Control (ACC) automatically adjusts a vehicle's speed to maintain a safe distance from the vehicle ahead, enhancing comfort and safety during varying traffic conditions [17].

[2]Lane Keeping Assist (LKA) actively helps to keep the vehicle centered within its lane, providing steering corrections if the vehicle begins to drift unintentionally, thus reducing the risk of lane departure accidents [18].



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There are two main problems with this MR: One is in the source scenario, where the event logic has not been clearly shown: the reason why the EGO vehicle does not drive into the parking space and how it relates to the parking function is unclear. The other problem lies in the follow-up scenario, in which the input relation is unclear (how the follow-up scenario has been derived from the source scenario) and the relation of the EGO vehicle's behavior between the source and follow-up scenario is not specified. Moreover, the description of the EGO vehicle's behavior lacks specificity in demonstrating the decision-making processes of the system. Instead, the expressions attribute human-like decision-making to the vehicle ("the EGO vehicle had better drive in the smaller spot"), which diverges from the expected output formats of an ADS, typically characterized by binary or clearly defined actions (e.g., proceed with parking maneuver or abort).

### *B. Applicability*

The applicability criterion received an average score of **3.5**, indicating that several MRs demonstrated potential for practical testing applications. One common problem with the generated MRs is that the description closely mirrors existing function specifications, making them more suitable for single-scenario testing [19] rather than emphasizing the relationships among different scenarios. However, there is a portion of the MRs demonstrates potential for testing applications. An example of MRs in this context involved manipulating the positioning of obstacles around the parking spot. Specifically, it proposed swapping the obstacles' locations on either side of the designated parking space, and the distance between the EGO vehicle and the nearby obstacles should be similar in both the source and follow-up scenarios.

### *C. Novelty*

The evaluation under the novelty criterion yielded an average score of **2.3**, indicating a general trend toward MRs with limited innovation. This observation aligns with the earlier discussion on applicability, where a portion of MRs were similar to functional specifications, thereby constraining their potential to provide new insights. Despite this trend, a few MRs stood out for their originality, such as an MR exploring the effects of relocating a light source around the parking spot to test the ADS's parking angle consistency. The MR is about placing the light ahead of the target spot in the source scenario and moving the light to the back of the target spot in the follow-up scenario, where the angle of the EGO vehicle when it parks into the slot should be similar in both scenarios. This MR differs from typical obstacle-based tests and is innovative in using the placement of lights to test the angle of the vehicle.

### *D. Clarity, Relevance to Safety, and Computational Feasibility*

These criteria each received an average score of **2.8**. A significant portion of the MRs suffered from ambiguous descriptions and lacked a focus on safety aspects. For instance, below is an example of the MRs generated:

TABLE II: Participants' feedback regarding ease of generating MRs

| Ease of Generating MRs | Percentage |
|---|---|
| Difficult: Had to overcome some challenges and occasionally sought guidance. | 17% |
| Moderate: encountered occasional challenges but managed to navigate through. | 33% |
| Easy: Found the process fairly straightforward with minimal challenges. | 33% |
| Very Easy: The process was intuitive and faced no significant challenges. | 17% |

TABLE III: Participants' feedback regarding the confidence of generated MRs

| Confidence in Generated MRs | Percentage |
|---|---|
| Low: Slightly unsure about certain aspects of the MRs. | 17% |
| Moderate: Fairly confident, but see areas of potential improvement. | 33% |
| High: Very confident in most parts, with minor uncertainties. | 33% |
| Very High: Extremely confident; certain about the quality and effectiveness of the MRs. | 17% |

> *"In the source scenario, the Ego car is going to drive into the parking lot with a lot of parking spots. In the follow-up scenario, suddenly, someone has fallen or tripped over a pack of fruit or other commodities left in your parking spot. The Ego vehicle stops and waits or drives away to find a new parking spot."*

This MR has shortcomings in three aspects: firstly, in the source scenario, only the initial behavior of the EGO vehicle is mentioned without detailing the ADS's interaction with the environment or other traffic participants. Secondly, the follow-up scenario's description of the EGO vehicle's action does not concretely outline the ADS's decision in response to the pedestrian hazard. The tester should specify a single action taken by the vehicle instead of presenting options such as 'stops and waits' or 'drive away', indicating a misunderstanding of the system. This absence of detail decreases the MR's utility in testing the ADS's safety mechanisms, and presents challenges for accurately simulating or implementing the MR in real-world tests. Thirdly, the connection between the behaviors of the EGO vehicle in both the source and follow-up scenarios is unclear, making it difficult to determine any violations of MRs during the experiments.

### *E. Overall Usefulness*

The overall usefulness criterion scored an average of **2.6**, reflecting a mixed assessment of the MRs generated by humans. This score reflects the challenges faced in the generation of MRs, with a notable number being either incomplete or inaccurately conceptualized, which consequently limits their practical application in testing environments. Despite these limitations, it is important to acknowledge that within the MRs evaluated, some MRs stood out for their innovative approach and relevance, demonstrating potential value for implementation in testing scenarios.



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## V. Discussion

### A. Comparsion of Students-generated and ChatGPT-generated MRs

In this study, we compared the qualities of MRs generated by humans, who were beginners in the area and received only forty minutes of instruction, with ChatGPT's generated MRs in the previous study [8], as Fig. 1 shows.

The clarity scores of MRs generated by human beginners and ChatGPT were similar. Both groups produced MRs that were easily understood but closely aligned with function specifications. This could have led to a lack of focus on the logical progression and connection between scenarios. Such connections are essential for MRs to effectively test system responses across different scenarios [3]. In terms of safety relevance, ChatGPT's MRs slightly outperformed those of the humans. This suggests that ChatGPT might be better at incorporating safety considerations into MRs than human novices.

ChatGPT's MRs better in applicability, computational feasibility, and overall usefulness may be due to its extensive training on diverse datasets [20], allowing it to generate more MRs that are well-tuned to existing specifications and systems. The slightly better correctness score for ChatGPT could be because it follows logical and real-world data patterns [20], an area where beginners might struggle without extensive knowledge or experience.

Notably, despite their brief education on the subject, the human participants managed to score higher than ChatGPT in the novelty criterion. This suggests that even at a beginner level, there is a chance for humans to approach problems with an innovative perspective, leading to generating testing scenarios that an AI may not consider: This is impressive, given the complexity of MR generation and the limited exposure to the subject matter.

The comparative analysis of the MRs suggests that while AI can provide technically sound and applicable MRs, the human capacity for innovation, particularly after only a brief exposure to the subject, is noticeable. This indicates a potential for growth and learning in human participants, which could be further realized with more extensive training.

### B. Interpretation

Tables II and III show the self-evaluation results of the students regarding the ease of generating MRs and their confidence in the MRs produced.

The self-evaluation on the ease of MR generation and confidence in the results offers valuable insights for teaching and practice. The diversity in the ease of MR generation suggests that while some students found the methodology accessible, there was a notable variation in how different students perceived the complexity of generating MRs. The fact that 17% found the process 'Difficult' points to a need for clearer and more detailed instructions that can cater to a range of learning styles and paces, ensuring that all students receive the support they need.

Confidence levels in the MRs produced indicate that while a substantial number of students felt confidence in their work, some students expressed reservations, with 17% indicating 'Low' confidence. This split underscores the importance of reinforcing confidence-building in MR generation through practice and feedback. It also highlights the potential for introducing peer review or collaborative learning exercises where students can engage with and learn from each other's approaches and thought processes.

The variation in both ease of MR generation and confidence levels indicates that while the workshop's one-hour timeframe was designed to maximize participation by not being overly time-consuming, it may have influenced the depth of understanding for some students. In an empirical study of exploring MT effectiveness [4], it was demonstrated that inexperienced students with a longer time of learning (three hours) could deliver a sufficient number of appropriate MRs. The decision to use a shorter time frame in the research was made to attract more participants and replicate situations where time is limited, like brief guest lectures, as longer sessions might discourage potential attendees. However, this decision may have implications for the thoroughness of the educational experience, suggesting a trade-off between attracting participants and providing extensive instruction. Meanwhile, the quality of MRs and the ease of generation process are also dependent on the complexity of the SUT. In this study, the ADS function might be relatively more complex than the SUTs that the students faced in the empirical study [4], which comprised five JAVA math programs.

Overall, these self-evaluations reveal that while the initial instruction provided a foundation, the teaching strategies could be further refined. Enhancing the clarity of instruction, offering more practical examples, and increasing interactive elements may help students overcome challenges and build greater confidence in their abilities to generate effective MRs.

### C. Limitations and Future Work

The design of this pilot study, with a one-hour session to attract participants, presents both strengths and challenges. While the length made the session more accessible, leading to a varied range of participants, it also limited the depth of instruction, as evidenced by the diverse levels of confidence and ease in generating MRs reported by the students. Some students indicated a need for further guidance, suggesting that additional time might enhance comprehension and skill acquisition.

The number of participants in this study was relatively small, which serves as an initial exploration into human performance in the MR generation. This participant size indicates the need for larger studies to confirm and expand upon our preliminary findings. Moreover, the efficacy of employing human testers or ChatGPT to generate MRs depends on various aspects, including the complexity of the system being evaluated, the accessible domain expertise, and the particular demands of the testing procedure.



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Fig. 1: Comparison of students-generated and ChatGPT-generated (GPT-3.5) MRs

Future workshops would consider lengthening their duration to better balance the need for participants' availability with the depth of learning provided. Moreover, peer reviews and collaborations could be introduced to reinforce the understanding and ability of MR generation among participants, particularly for those who expressed lower confidence levels or found the process challenging. Furthermore, the variety of the SUT and the participants' varying levels of domain knowledge underscore the importance of tailored educational strategies in future research.

## VI. Conclusion

This paper presented a pilot study on the ability of novice testers to generate MRs for a typical ADS function, assessing not only the learning outcomes but also comparing these human-generated MRs against those produced by AI. The results demonstrate that, even with minimal instruction, participants were able to create MRs with a wide range of correctness, applicability, novelty, and usefulness. A key observation is the human participants' capability for innovative thought, contrasting with AI's strength in generating MRs that excel in technical precision and applicability. Future work will aim at enhancing educational approaches to improve both the quality of MRs produced by humans and their confidence in the MRs that they generate, even with limited training.

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